

ALAA SHAMMOUT

Dubai, UAE | +971 50 759 3999 | AlaaShammout03@gmail.com

<https://www.linkedin.com/in/alaa-shammout03/> | <https://github.com/AlaaShammout03> | [portfolio url](#)

EDUCATION

American University of Sharjah

Bachelor of Science in Computer Engineering

Sharjah, UAE

Expected: June 2026

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Cloud Computing, Computer Vision, Embedded & Real-Time Systems, Wireless Networks, Advanced Digital System Design.

TECHNICAL EXPERIENCE & PROJECTS

Senior Design Project - Lead Frontend Developer (Sponsored by Emagine Solutions)

Sep 2025 - May 2026

4-person engineering team | AI-powered EV Charging Support & Maintenance Platform

- Led end-to-end frontend development for two React/Tailwind applications: a customer-facing web app and an operator dashboard.
- Built a live operator dashboard featuring real-time telemetry charts, AI-classified ticket management with confidence scores, predictive maintenance alerts, and a fleet incident mode that activates dynamically on threshold breach.
- Implemented a Smart Route feature using the Mapbox Directions API with live traffic routing, suggesting underutilized chargers to users with estimated drive/wait/charge breakdowns and a points-based incentive system.
- Integrated the frontend with a FastAPI/PostgreSQL backend, including JWT authentication, OTP login, ML classification results, IoT telemetry streams, and dozens of REST API endpoints.

Crowd Counting with Deep Learning - Computer Vision Project

Spring 2026

- Re-implemented CSRNet for single-image crowd counting on the ShanghaiTech benchmark, achieving MAE of 18.01 on Part B and 106.11 on Part A.
- Improved density-regression training stability by addressing optimizer-related instability and applying density rescaling to stabilize gradient behavior.
- Conducted per-density-bucket error analysis to compare model performance across low-, medium-, and high-density crowd scenes.

Cloud-Native Booking Microservice on Azure - Smart Urban Platform

Spring 2026

- Built a Spring Boot REST booking microservice for a smart-city resource reservation platform, supporting reservation creation, rescheduling, cancellation, and user booking dashboards.
- Implemented REST APIs backed by PostgreSQL and Spring Data JPA, including active bookings, booking history, booking limits, and conflict detection.
- Added operating-hours validation, bookable-resource checks, and transactional PostgreSQL advisory locking to reduce double-booking risk.

FreeRTOS Multi-Stage Embedded Game on ESP32 - Embedded System Project

Spring 2026

- Built a multi-stage embedded “escape room” game on ESP32 using FreeRTOS, integrating keypad, joystick, RGB LEDs, buzzer, and I2C LCD components.
- Implemented concurrent tasks with mutexes and semaphores to coordinate input handling, game logic, display updates, and hardware feedback.
- Validated the system in Wokwi and documented the architecture with task diagrams, synchronization tables, and a stage by stage walkthrough.

TECHNICAL SKILLS

Backend, Cloud & Databases: Spring Boot, FastAPI, REST APIs, PostgreSQL, Spring Data JPA, Microsoft Azure

Frontend: React.js, Tailwind CSS, HTML, CSS

Embedded & Digital Design: C, C++, FreeRTOS, ESP32, Verilog, Intel Quartus, Wokwi

ML & Vision: Python, PyTorch, OpenCV, NumPy, pandas

Tools: Git, Linux/Bash, VS Code

Spoken Languages: Arabic (Native), English (Fluent), Turkish (Intermediate)

LEADERSHIP & ACTIVITIES

- AUS Clubs (2021 - 2026): Active in the AI Club, IEEE EMBS, and Computer Engineering Club - technical workshops and peer mentoring.
- Competitive Athletics (2018 - Present): Bronze medalist, AUS Men's Swimming (2022); university football and boxing.